Build or Burn: A Return Term for Government Spending in the Inflation Equation

Abstract

Government spending feeds inflation or growth according to what it returns, because a dollar spent adds demand and adds supply only through the goods and services it delivers. The national accounts value government purchases at cost, as though every public dollar returned one dollar of value, and the quantity equation, MV = PQ, leaves unexplained where the Q comes from. This paper adds the missing term. It calls the approach BERN, Bernstein Economics of Realized Net: judge whether every dollar a government spends feeds inflation or growth by the competitively priced goods, services and national skills it delivers. Is it a good investment? Does it build or burn?

Effective government spending is spending weighted by its realized return, and the price level is demand divided by capacity. Spending is inflationary when it adds demand faster than the capacity it delivers, and it is not when its return keeps pace. The adjustment puts the four rival accounts of money and prices into one equation: the quantity theory describes spending that returns nothing, the Keynesian multiplier describes spending into idle capacity, the Austrian cycle describes investment whose return falls short, and Modern Monetary Theory's resource limit is the point where the return term decides. The hyperinflation record sits at the burn extreme: money-financed spending soared while production collapsed. Spending that builds competitively priced capacity also draws demand from the rest of the world, which supports the currency. The paper states the condition, works it through a two-period example and proposes a test.

Keywords: government spending; inflation; quantity theory of money; national accounts; public investment; Modern Monetary Theory; hyperinflation; return on investment

1. The missing term

Output equals consumption plus investment plus government purchases plus net exports (exports minus imports), Y = C + I + G + NX. The identity is true by construction. It counts each public dollar at what it cost, because most of what a government provides has no market price, and the national accounts (the official statistics measuring a nation's output, income and spending) value it by the inputs used to make it (United Nations et al., 2009). That convention counts every public dollar as a dollar of output, whatever it returns. A bridge that carries freight for eighty years and a bridge to a village of four hundred people that already has a ferry enter the accounts as the same G. Valuing an output by its inputs is the old cost-of-production view of value, whose strongest form was the labor theory of value (Ricardo, 1817; Marx, 1867). The marginal revolution of the 1870s replaced it with value at the margin, what buyers will pay for one more unit (Jevons, 1871; Menger, 1871), and BERN makes the same correction for public spending.

A private firm has the missing column. It is the return on invested capital: borrowing at five percent to earn three is a loss on a schedule anyone can calculate in advance. A government that spends money it creates faces the same arithmetic in another form. The money it adds is a claim on goods. If the spending produces the goods, the claims meet them. If it does not, the claims bid up the prices of the goods there are.

The equation of exchange, MV = PQ, makes the same point from the side of money (Fisher, 1911). The money spent, M times its velocity V (the rate at which money changes hands), equals the price level P times real output Q (the quantity of goods and services produced). It is an identity, and it is silent on what makes Q

grow. This paper supplies that link for public spending: the part of Q that government spending creates depends on the return the spending realizes.

2. The adjustment

2.1 Effective government spending

Count each outlay by what it delivers, not by what it cost. A road that carries trucks for decades returns more than it cost. A fireworks show returns one evening. Call the ratio of value delivered to cost the realized return. The national accounts set it to one for every outlay; BERN uses the realized value.

Stated precisely, for outlays G_i with realized returns R_i, effective government spending is

$$G^{*} = \sum_{i} R_i \, G_i,$$

where R_i is the real value of the goods, services and skills outlay i delivers over its life, at the prices buyers would pay for them, discounted to the date of spending (value arriving later counted for less), divided by its cost. An outlay builds when R_i > 1 and burns when R_i < 1, and the burn of a program is the value its spending failed to return, the sum of (1 - R_i) G_i over its burning outlays. Value that markets do not price, such as clean water, safe roads and working courts, counts at its appraised value, as cost-benefit practice already estimates it (HM Treasury, 2026).

2.2 Prices are demand over capacity

Prices rise when the money chasing goods grows faster than the goods. Spending adds money demand at once. It adds goods only when what it buys starts producing, and only in proportion to its return. So the same spending

pushes prices up during construction, and afterward prices settle back where the return is high and stay up where it is low.

Stated precisely, in an economy at capacity the price level is nominal demand (spending measured in money) over real capacity,

$$P_t = \frac{D_t}{C_t},$$

which is the quantity equation with D = MV and C = Q at full use. Prices then rise exactly when demand grows faster than capacity, and inflation is close to the growth of demand minus the growth of capacity. An outlay raises demand by its cost while it is spent and raises capacity by the output it delivers from its delivery date on. Spending is non-inflationary once delivered when the capacity it adds grows at least as fast as the demand it adds:

$$\frac{\Delta C}{C} \geq \frac{\Delta D}{D}.$$

The higher the realized return, the more of the spending's price effect its output cancels. Spending whose added capacity keeps pace with the demand it adds leaves no lasting rise in prices; spending that returns nothing leaves all of it, as demand with no goods behind it. Money created to pay for spending stays in circulation, so its output has to keep coming: a one-time injection that raises the money stock by a tenth needs capacity a tenth larger, year after year, to leave prices where they were. With idle capacity, C is not yet binding (not yet the limit on output), and spending raises output with little effect on prices until the slack is used.

2.3 A two-period example

With the money spent held fixed and only what it delivers varied, two projects push prices up together while they are built and separate once one of them

produces. Consider projects A and B with the same cost and the same spending path, each in its own simple economy with one composite good (a single good standing for all output), flexible prices and fully used capacity. Let D_t be nominal spending on the good and C_t the quantity produced at capacity, so that $P_t = D_t/C_t$. Both economies start at $D_0 = C_0 = 100$ and $P_0 = 1$. During construction, spending rises to 110 while capacity stays at 100. In the next period spending stays at 110; project A then raises capacity to 110, and project B adds nothing to this market.

Period	Spending, both	Capacity A	Capacity B	Prices A / B
Before construction	100	100	100	1.00 / 1.00
Construction	110	100	100	1.10 / 1.10
After delivery	110	110	100	1.00 / 1.10

A builds and B burns. Both raise prices by ten percent while they are spent. A's delivered capacity brings its price level back to where it started; B's stays ten percent higher. With the same money spent, the price levels stand in the inverse ratio of the capacity each economy ends with, $P_t^A/P_t^B = C_t^B/C_t^A$. The numbers are illustrative, and the mechanism is general: the inflation an outlay leaves behind is set by what it returns.

3. Four schools in one equation

The rival accounts of money and prices each describe one regime of the same equation, and the return term, together with the amount of slack, says which regime holds.

The quantity theory, in Irving Fisher's form (Fisher, 1911), holds where spending returns nothing and capacity is fixed: demand grows, capacity does not, and prices rise in step with money. That is the burn regime.

The multiplier of John Maynard Keynes (Keynes, 1936), the rise in output per dollar of new spending, holds where capacity stands idle: spending raises output with little pressure on prices, because C is not binding. The return term matters less while the slack lasts and decides the outcome once it is used up.

The Austrian account of Friedrich Hayek (Hayek, 1931) holds where credit finances investment whose realized return falls short of what its prices promised: capacity is built that does not pay back, and the bust is that shortfall coming due.

Modern Monetary Theory, set out by L. Randall Wray and Stephanie Kelton (Wray, 2012; Kelton, 2020), holds that a government issuing its own currency cannot be forced into nominal default, and that its real limit is the economy's capacity to produce. Both claims fit the equation. What MMT does not supply is a test for which spending relaxes that limit and which tightens it. The return term is that test: spending with a high realized return raises the ceiling it is measured against, and spending with a low one consumes the same resources and raises nothing.

4. Hyperinflation: the burn extreme

The hyperinflation record shows the burn regime at its limit. Phillip Cagan defined hyperinflation as prices rising more than fifty percent a month and studied seven European cases (Cagan, 1956). Steve Hanke and Nicholas Krus later catalogued 56 episodes (Hanke and Krus, 2013). The episodes share one pattern: very large deficits paid for by printing money, while production fell. Weimar Germany in 1923, Hungary in 1946, which holds the record, Zimbabwe in 2008 and Venezuela from 2016 all fit it. In Zimbabwe, the seizure of commercial farms cut farm output; in Venezuela, oil output fell by more than half while the central bank financed the state.

In the terms of this paper, these were economies where demand was pumped up with money while capacity shrank: D soared and C collapsed, so P exploded. The spending that drove them returned little or nothing. It paid for consumption, transfers (payments like benefits that buy no goods or services), losses and wars, and the output that would have met the new money was not built. Thomas Sargent found that four of the classic hyperinflations ended abruptly when governments changed the fiscal regime (the settled policy on taxing, spending and borrowing) that financed them (Sargent, 1982). Ending the burn ended the inflation.

The same record does not support a rule that any large deficit or debt ratio (government debt as a share of national output) must end in hyperinflation. High debt has coexisted with low inflation. What the money bought, and what it returned, separates the cases.

5. Build: good investment and the rest of the world

Spending that builds competitively priced goods, services and national skills does more than add domestic supply. Buyers abroad want what it produces. Export earnings pay for imports, which add to what domestic money can buy, and foreign buyers who need the currency to pay hold up its value. A good investment can therefore be paid for with new money and still leave prices steady, because the goods it returns meet the money and the rest of the world takes part of both.

Stated precisely, in an open economy the goods available to domestic buyers are domestic capacity plus imports, and imports are paid for by exports. Raising the competitive output of the economy raises exports, which finance imports without drawing down reserves (the foreign currency the central bank holds), and foreign demand for the currency lowers the velocity of the money held at home. This is why the realized return in Section 2 counts value at the prices buyers anywhere

would pay: a project whose output no one would buy at its cost returns less than its accounts claim, however it is classified. Evidence on public investment points the same way: its output effects are larger where investment efficiency is higher (International Monetary Fund, 2014), and implementation delays shift its effects later (Leeper, Walker and Yang, 2010).

6. Measuring the return

The return is measurable with tools that already exist. Transport agencies compute benefit-cost ratios as routine, and HM Treasury's Green Book sets out how to appraise the costs, benefits and risks of public options, including benefits with no market price (HM Treasury, 2026). After the Atkinson Review, the United Kingdom began measuring the output of some public services directly instead of assuming it equals their cost (Atkinson, 2005). BERN asks for the step that is usually missing: check the return after delivery, report it beside the spending with its uncertainty, and use the realized figure, not the projected one, in the inflation account. If the traffic did not come, the finished bridge should not keep its projected return.

7. Predictions and a test

Three predictions follow. Given the same money growth, inflation should be lower where a larger share of spending builds, after a lag equal to the time its projects take to deliver. Hyperinflation should require the burn regime: money-financed spending with little return while capacity falls. Spending whose delivered capacity keeps pace with the demand its money adds can be financed with new money without lasting inflation, and the price rise during construction should reverse.

A test can use project-level data with delivery dates and ex post returns (returns measured after the fact), such as the World Bank's independent evaluations,

merged with money, output and price data. It should compare episodes of similar money growth with different shares of building and burning spending, and use differences in project timing that were not chosen in response to inflation. Statistical controls (adjusting for other measured factors) alone do not establish the mechanism, and no such study is reported here.

8. Objections

"An identity cannot be wrong, and adding a return term breaks it." The expenditure identity stays as it is: G is what was spent. BERN adds a second account, effective spending, and an equation for what makes capacity grow. The identity says spending equals spending; BERN says what the spending returns.

"The return is too hard to measure." It is estimated before most large projects are approved. BERN asks that the estimate be checked afterward, reported with its range and carried into the inflation account. A misleading number is worse than none, so the assumptions, the alternatives and the ranges should be published.

"Some public goods have no market return." The realized return counts value that markets do not price, at its appraised value. A courthouse, a vaccination campaign or a working legal system returns real value, and the method values it as appraisal already does.

"Financing, expectations and crowding out matter too." They do, and the equation keeps them: financing and expectations act through demand and velocity, and crowding out through the capacity other uses lose. BERN isolates the term the standard accounts leave out, what the spending returns.

9. Conclusion

Government spending feeds inflation or growth according to what it returns. The national accounts count every public dollar as returning one dollar, and the quantity equation leaves the growth of output unexplained. BERN replaces the assumed return with the realized one: effective spending is spending weighted by its return, and prices are demand over capacity. Spending that builds competitively priced goods, services and skills meets its money with goods and draws in the rest of the world. Spending that burns leaves its money chasing goods nobody made, and at its extreme it is the hyperinflation record. The question for every dollar is the one a firm asks of every investment: does it build or burn?

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